A 3D Simulation Platform for Fuel Handling and Storage Systems

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Abstract. The Fuel Handling and Storage System (designation code: PMC) is a critical subsystem in nuclear power plants, responsible for the safe loading, transfer, storage, and retrieval of fuel and core components during refueling. Conventional PMC systems rely heavily on manual visual inspection, and while 2D monitoring based on sensor signals offers supplementary support, it lacks integration of multi-device, multi-viewpoint, and cross-temporal information, limiting safety and efficiency. We present a 3D simulation platform, built with a industrial quasi-digital twin strategy, that combines high-fidelity modeling with spatiotemporal data from industrial encoders to reconstruct and synchronize equipment states. A dedicated hardware–software architecture and control algorithms convert operational data into unified flows driving real-time simulation and 3D rendering. The visualization results are delivered to multiple touchscreen terminals, offering interactive feedback and user-controlled viewpoints that improve coordination among personnel and equipment. The platform enhances information clarity, spatial awareness, and interference or collision avoidance compared with conventional systems. Besides online simulation, the platform also supports offline simulations using historical logs and procedural files, enabling repeated training and reducing potential operational errors in live environments. A built-in spatial mapping algorithm adapts the virtual environment to plants of varying scales. Validation via gripper vibration evaluation and system error analysis confirms spatiotemporal consistency and high simulation accuracy between real and virtual environments, providing a practical foundation for future PMC enhancements.

Keywords: 3D Simulation · Fuel Handling and Storage · Industrial Digital Twin · Interactive Visualization.

1 Introduction

The Fuel Handling and Storage System (designation code: PMC, also referred to as RFH in certain updated nomenclature) is a mission-critical subsystem in

nuclear power plants, responsible for the secure receipt, transfer, storage, and retrieval of nuclear fuel and core components during reactor refueling [1]. In current practice, three major devices form the operational core of the PMC: the spent fuel pit crane (SPC), the fuel transfer device (FTD), and the manipulator crane (MC). These components work in sequence to move fuel assemblies between the spent fuel pit, transfer pit, and core pit, strictly following predefined operational procedures. The operational environment imposes stringent constraints. Fuel assemblies must be handled with high positional accuracy under complex conditions involving path planning, mechanical characteristics of the handling devices, and segmented work areas distributed across separate plant buildings to minimize radiation exposure and account for elevated water temperatures. Many tasks take place underwater, often requiring synchronized actions between teams operating in physically isolated zones.

Conventional PMC implementations still rely heavily on manual visual judgment to tolerate positional deviations. Although 2D graphical interfaces based on sensor feedback are used as auxiliary aids, they fail to provide an integrated operational picture that fuses multi-device, multi-view, and cross-temporal information. End-to-end refueling procedures often lack embedded spatiotemporal path-planning capabilities, making it difficult to foresee mechanical interference or procedural conflicts. Furthermore, the underwater working environment restricts the ability to inspect potential collisions from multiple angles in real time, which constrains operational safety. Current procedure validation is often limited to simplified, single-building simulations, reducing applicability to plants with differing structural layouts. The main limitations can be summarized as: (1) fragmented real-time data from independent devices remain unintegrated; (2) operational verification relies on procedural documents and operator experience, with procedure validation rarely accounting for structural variations across facilities; (3) the absence of multi-view, collaboratively controllable, interactive monitoring of equipment status and environmental conditions makes it difficult to detect potential issues in time.

To address these challenges, we develop a PMC 3D Simulation Platform that integrates a complete hardware–software architecture with a industrial quasi-digital twin (DT) strategy. The platform combines detailed 3D models with spatiotemporal data streams from industrial PLC-connected encoders. These measurements are converted into unified control instructions that drive real-time simulation and rendering. In addition to online real-time visualization, the platform offers offline simulation using archived historical operation logs and procedural files, enabling operators to rehearse complex maneuvers and repeat training without disrupting live operations. A suite of control algorithms, including sensor data dropout mitigation to stabilize communication between hardware and software layers, a heuristic early-warning mechanism derived from on-site motion data for collision prediction, and an adaptive speed control model, ensures the robustness of both online and offline simulations. To support deployment in plants with varying layouts, the system incorporates retargeting (or redirection) [6,19,11] techniques from virtual reality (VR) and a multi-stage spatial map-

ping algorithm that adapts the unified virtual workspace to facilities of different scales. Simulation results are delivered to multiple interactive touchscreen terminals, enabling adjustable viewpoints and coordinated operation among multiple users and devices. This design improves spatial clarity, strengthens situational awareness, and provides actionable feedback beyond the capabilities of traditional systems.

In summary, the main contributions of this work are:

- A complete hardware–software architecture for PMC 3D simulation, inspired by a industrial quasi-DT strategy;
- A suite of control algorithms enhancing simulation robustness, precision, adaptability, and portability;
- A multi-user, touch-enabled 3D visualization interface designed for collaborative fuel-handling and storage operations.

2 Project Background

This section introduces the related work of the project and outlines the physical structure of the PMC on which the 3D simulation platform is modeled.

2.1 Related Work

Digitalization technologies have been increasingly adopted in nuclear power plant subsystems to enhance safety, operational efficiency, and training effectiveness. As the International Atomic Energy Agency (IAEA) has noted, beyond the primary power generation systems, auxiliary subsystems such as the Chemical and Volume Control System [12] and the Reactor Cavity and Spent Fuel Pit Cooling and Treatment System [21] have been early targets for digital integration. These developments reflect a broader trend toward incorporating real-time sensing, advanced simulation, and intelligent control into plant-wide digital infrastructures. Fuel assembly handling, transfer, and storage are core functions of the PMC, ensuring the safe movement of fuel between the spent fuel pit, transfer pit, and reactor core pit [9]. The IAEA has emphasized that digital tools, VR, and robotic systems can accelerate tasks such as fuel unloading and facility decommissioning while improving coordination and safety [10]. Industrial practices increasingly deploy 3D simulation, VR visualization [3], and DT [20] architectures to optimize refueling workflows, support multi-equipment collaboration, and enable predictive fault detection [2,16]. Recent research has advanced DT solutions for nuclear systems [18,14], addressing both physics-informed and data-driven modeling. Examples include reduced-order modeling for reactor physics [7] and hybrid algorithms for parameter identification and state estimation in reactor operation DTs [8]. Other studies have developed hybrid physics– and data-driven DT models for nuclear power plants [4], while prognostic DTs have been explored for auxiliary systems such as NAMAC [13]. Specific DT applications include environmental assisted fatigue assessment for plant components [5]. These

works highlight the adaptability of DT frameworks to diverse operational contexts, ranging from heavy mechanical handling to structural health monitoring. The educational and training dimension of VR is also gaining recognition. The Nuclear Institute's Nuclear Reactor Simulator [15] and the IAEA's VR-based research reactor models [17] demonstrate the utility of immersive simulations for both operator training and safeguards education. Scenario-driven VR environments support experiential learning, promote procedural compliance, and can be potentially coupled with DT backends for realistic physics behavior.

2.2 Structural Overview of the Facility

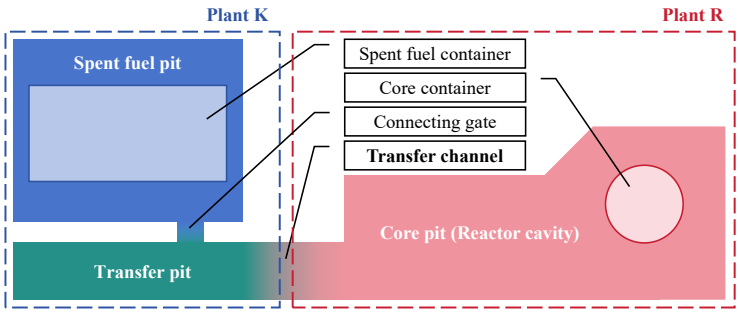


Fig. 1. Abstracted physical layout of PMC-related plant facilities, illustrating the spatial arrangement of three pits across Plant K and Plant R.

The PMC system is designed to handle nuclear fuel assemblies during reactor refueling outages, with a focus on safe and efficient fuel handling and storage operations. As shown in Figure 1, the overall layout consists of multiple interconnected facility zones that operate in coordination, often with underwater interfaces. Structurally, the system can be abstracted into three primary pits: (1) **Spent fuel pit** – equipped with racked containers for storing non-fissioning (spent or fresh) fuel assemblies; (2) **Core pit** (reactor cavity) – housing the core container (pressure vessel) where nuclear fission reactions occur during plant operation; (3) **Transfer pit** – serving as an intermediate connection between the two. To ensure radiation shielding and operational safety, the spent fuel pit and transfer pit are located in the same reinforced-concrete building, designated as Plant K, and are linked internally by a *connecting gate*, which allows controlled transfer of fuel between the two areas. The core pit is located in a separate reinforced-concrete building, Plant R. The two buildings are joined indirectly via a *transfer channel* extending from the end of the transfer pit.

Figure 2 illustrates the primary handling equipment. Operations in Plant K are mainly performed by: (1) the **spent fuel pit crane** (SPC), while operations in Plant R are handled by (2) the **manipulator crane** (MC). The two buildings are connected by (3) the **fuel transfer device** (FTD), which transports fuel

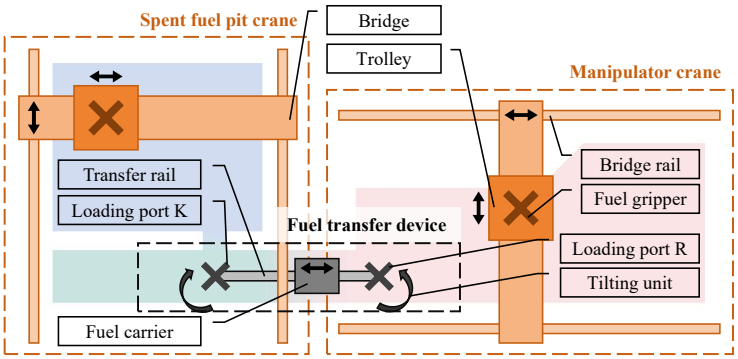


Fig. 2. Abstracted mobility and coordination of PMC-related handling equipment, showing the operational ranges and cooperative workflow of the SPC, FTD, and MC.

between the pits along the transfer channel. Each crane comprises three main components: the *bridge* (large cart), which travels along ceiling-mounted *bridge rails*; the *trolley* (small cart), which moves perpendicularly along the bridge; and the *fuel gripper*, which raises and lowers vertically to engage the fuel assemblies. In the diagram, the vertical position of the gripper is indicated by a dark orange "X." In the SPC, the gripper is suspended by pulleys, steel cables, and hooks, with on-site personnel applying manual damping to prevent excessive swing. The MC instead uses a rigid telescopic sleeve and a pneumatically actuated gripper, enabling precise vertical movement and secure handling during core operations. The FTD consists mainly of a *tilting unit* and a *fuel carrier*. During loading, the SPC delivers a fuel assembly to the tilting unit at *Loading Port K*, marked with a dark gray "X," which is initially vertical. Once positioned, the tilting unit rotates 90° to align the assembly horizontally under controlled hoist slackening (located above). The fuel carrier then moves the assembly along the *transfer rail* to *Loading Port R*, where the hoist tightens to return the tilting unit to a vertical position. The MC then retrieves the assembly for core loading. Unloading follows the same sequence in reverse.

3 Platform Architecture Design

This section presents the detailed architectural hardware-software design of the 3D simulation platform, along with its embedded control algorithms.

3.1 Hardware Architecture

Taking the facility layout described in subsection 2.2 as a reference, the hardware architecture shown in Figure 3 is designed to capture spatiotemporal signals from the field, track the status of equipment and fuel assemblies across Plants K, R, and the transfer channel, and stream data to the rendering terminals.

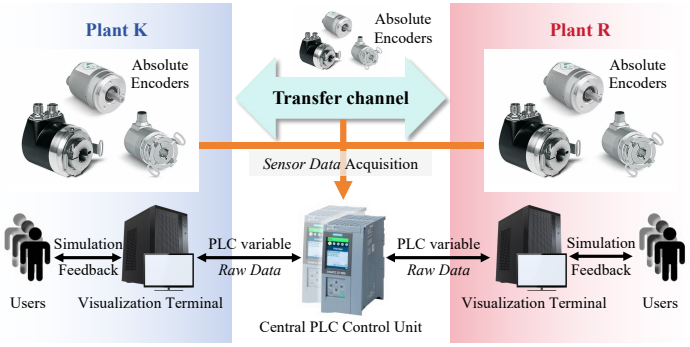
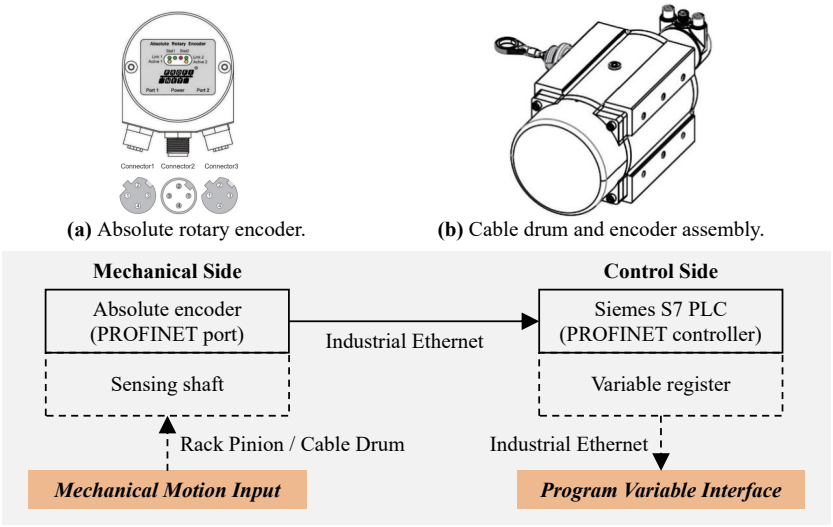


Fig. 3. Overview of the hardware architecture. The central PLC control unit functions as the core hub, relaying raw data from absolute encoders deployed in Plant K and Plant R to the visualization terminals. This real-time data acquisition chain enables simulation feedback to be presented to end users for interactive monitoring and control.

Absolute rotary encoders (see Figure 4) with PROFINET-IO RT interfaces are installed on key equipment motion axes across Plants K, R, and the transfer channel. Enclosed in stainless-steel housings, these encoders provide 13-bit single-turn and 14-bit multi-turn resolution, with repeatability within ± 1 bit. Their support for real-time industrial Ethernet communication makes them well-suited for continuous position monitoring in nuclear fuel handling. Encoder configurations are selected to match the mechanical characteristics of each subsystem. Bridge and trolley motions in both cranes use rack-and-pinion–driven encoders, which convert linear displacement into rotary motion and are ideal for high-precision linear travel measurement. Main hoisting systems and tilting mechanisms use cable-drum–driven encoders, providing a measurement precision of $\pm 0.05\%$, travel ranges up to 15 m, and operating speeds up to 6 m/s. This configuration enables compact vertical measurement assemblies while supporting long-travel applications.

For safety and reliability, all critical axes employ dual-redundant encoders, with a differential comparison mechanism to detect faults. In the bridge drive, dual encoders are mounted on one side; when their readings differ beyond a set threshold, an alarm is raised but movement continues, allowing operators to switch to the backup channel. If both sides fail, the bridge is brought to a stop. For the trolley, a threshold exceedance triggers an immediate halt, again with backend switching capability. For the main hoist and tilting winch, any excessive discrepancy between the two encoders causes the axis to stop, after which operation can be resumed via backend channel switching.

The PLC control system continuously samples encoder values to perform closed-loop monitoring. For horizontal motion (bridge and trolley), measurement uncertainty is maintained within ± 0.5 mm, with absolute positional accuracy better than 1 mm. For vertical motion (hoisting and tilting), the system achieves an overall control accuracy within ± 5 mm, meeting the strict displace-



(a) Absolute rotary encoder. **(b)** Cable drum and encoder assembly.

(c) Signal flow diagram from mechanical motion to PLC variable interface.

Fig. 4. PROFINET-based integration of absolute encoders with the facility front-end and the central PLC core hub.

ment tolerances required for nuclear fuel handling. As shown in Figure 5, all encoder data are transmitted via PROFINET industrial Ethernet to a dual-redundant PLC control unit, and then mapped into program variable tables for use by the graphical workstation. This workstation drives the 3D simulation rendering and user interaction terminals, forming a closed-loop chain of *sensing* $\rightarrow$ *control* $\rightarrow$ *simulation* $\rightarrow$ *interaction* that improves process visibility, fault tolerance, and safety margins during both live operations and training.

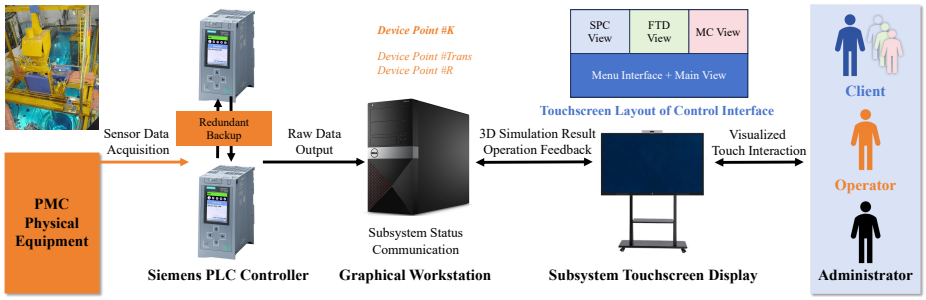


Fig. 5. Macro-level data flow and the hardware design for a representative device point.

Figure 5 illustrates the downstream hardware configuration for a representative equipment point under encoder integration. Taking Plant K as an exam-

ple, the graphical workstation reads the PLC variable table, processes the data through the software system (subsection 3.2), and generates an interactive 3D simulation view. The output is presented on a front-end touchscreen that supports multi-level user interaction. The PLC system is based on Siemens S7-1500 series 4 MB ROM controllers in a dual-redundancy setup. The graphical workstation is equipped with an Intel i9-12900 CPU @ 3.90 GHz, 32 GB RAM, and an RTX 3080 Ti GPU. The front-end display is a 55-inch 4K (16:9) multi-touch screen, enabling high-fidelity visualization and responsive multi-user interaction.

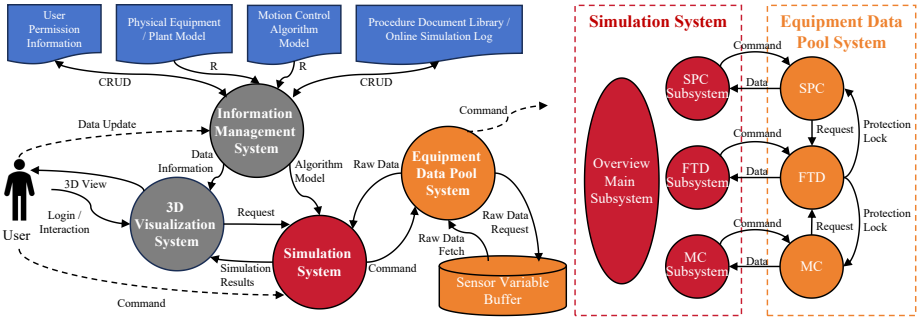


Fig. 6. Software architecture of the PMC 3D simulation platform. The left diagram illustrates the interactions among the four core systems. The right diagram details the internal subsystem structure of the Simulation System, showing bidirectional command–data exchanges with the corresponding physical devices via the Data Pool.

3.2 Software Architecture

The software architecture of the platform, illustrated in Figure 6, is organized into four core systems: the **Information Management System**, the **Equipment Data Pool System**, the **Simulation System**, and the **3D Visualization System**. These adopt a modular design and operate in close coordination to enable quasi-DT modeling of PMC equipment operating states, command dispatch, simulation-driven reasoning, and interactive 3D visualization.

Information Management System Situated at the upstream end of the software workflow, the Information Management System provides unified ingestion, storage, scheduling, and updating of all structured data. It interfaces with multiple data sources, including: (1) User permissions and role definitions; (2) PMC plant structure and equipment models; (3) Motion control algorithms and logic models; (4) Online simulation historical operation logs and procedural file repositories. The system integrates permission data, geometric models, and control rules to supply the Simulation and Visualization subsystems with the necessary

datasets, while also archiving simulation results from repositories. Administrators can perform full CRUD (create, read, update, delete) operations on user permissions, simulation models, logic rules, and scheduling plans. Operators can manage equipment status. Regular users, in accordance with their access rights, can browse or execute procedural files and review historical logs.

Equipment Data Pool System The Equipment Data Pool System functions as an intermediary buffer and abstraction layer between the Simulation System and on-site equipment, providing a unified interface for accessing equipment information. Leveraging the PLC Ethernet interface described in subsection 3.1, it uses TCP/IP to query sensor variable buffer addresses within the local network, polling at 20 Hz to continuously retrieve raw data. These data include encoder positions, motion feedback, load measurements, and deflection angles of on-site devices. Once acquired, the data are normalized and forwarded to the Simulation System for quasi-DT computations. The Data Pool supports concurrent requests and includes packet-loss mitigation (see section 3.3) to ensure stable, real-time acquisition. It also supports reverse command dispatch (for example, sending emergency stop signals) by relaying alerts or anomaly-based stop commands from the Simulation System back to the equipment control layer. This capability forms the foundation for a preliminary "virtual–physical closed loop" for coordinated control.

Simulation System The Simulation System is the computational core of the platform, supporting two primary operational modes:

- **Online Simulation**: Reads real-time sensor variables from the Equipment Data Pool to maintain synchronous DT simulations with the physical PMC system. It supports real-time motion computation, rendering, and state prediction, along with command dispatch, emergency stop handling, and simulation log recording for later analysis and traceability.
- **Offline Simulation**: Loads either (1) historical operation logs or (2) procedural files to reconstruct simulation sequences through playback, or uses algorithmic models to predict the outcomes of alternative motion paths. This mode is valuable for operator training, scenario validation, and virtual prototyping. Operation logs share the same raw data format as online simulations, while procedural files define end-to-end path sequences used by nuclear power operators for refueling planning or training.

To enhance modularity and reuse, the Simulation System is organized into four subsystems: (1) **Main Overview Subsystem** handles global simulation modeling, main loop control, time progression, module scheduling, and spatial alignment (section 3.3); (2) **SPC Subsystem** manages the lifting structures and motion logic of the Plant K spent fuel pit crane, including the bridge, trolley, hoist, and encoder components visualization; (3) **FTD Subsystem** models motion control for the transfer channel and associated devices, including transportation paths and posture changes of fuel assemblies between the two plants,

while monitoring or simulating protection locks to avoid conflicts between the bridge crane and the fuel handling machine; (4) **MC Subsystem** simulates the telescoping, loading, and unloading operations of the Plant R manipulator crane, ensuring virtual–physical consistency at critical reactor core locations. Each subsystem can operate independently for local testing or be integrated for full-system simulation.

3D Visualization System The 3D Visualization System serves as the primary interface for user interaction with the Simulation System, offering multiple view modes, parameter displays, and an interactive dashboard. Users log in via a touch-enabled terminal and operate within the privileges granted to their role:

- **Regular Users**: View 3D scenes, observe simulation processes, adjust viewpoints to inspect equipment states, and monitor parameters and trends.
- **Operators**: In addition to the above, perform interactive simulation control, load and import logs or procedural files, edit training paths, trigger emergency stops in online mode, and start or pause offline simulations.
- **Administrators**: In addition to the above, update datasets, manage user accounts, and modify simulation configurations.

The layout follows the four-section dashboard shown in Figure 5, corresponding to the four Simulation System subsystems. It supports real-time rendering, multi-user synchronized interactions, and automated camera tracking of points of interest. Users can search and highlight specific PMC components, trigger close-up camera views, and preview disassembly animations, providing an intuitive and immersive interaction experience.

3.3 System Control Algorithms

Sensor Data Dropout Mitigation Short network glitches or PLC jitter can momentarily drop encoder packets, causing state desynchronization or visual popping in the simulation. We use a lightweight hybrid correction that blends the latest valid measurement with constant-velocity extrapolation from recent frames. Let $\delta_{\eta_{i+1}} \in \{0, 1\}$ indicate whether sensor η returns a value at step $i{+}1$; φ is the inferred parameter and ω the measured parameter:

$$\varphi_{i+1} = \delta_{\eta_{i+1}}\, \omega_{i+1} + \big(1 - \delta_{\eta_{i+1}}\big)\left(\varphi_i + \frac{\varphi_i - \varphi_{i-1}}{\Delta t_i}\, \Delta t_{i+1}\right). \tag{1}$$

When a packet arrives (δ=1), the stream is snapped to the measured update parameter. Otherwise, a smooth prediction is produced by linear extrapolation from the last two frames. This keeps the virtual and physical timelines aligned and suppresses rendering jumps without introducing model-specific parameters.

Collision Detection In Plant K, the SPC gripper is cable-suspended (four cables, pulleys, hook, gripper). When the trolley brakes from maximum speed to rest, the gripper exhibits post-stop swing whose trajectory is well approximated by a Lissajous-like ellipse. In the ideal undamped case with identical initial kinetic energy, the swing-end amplitude δ scales with the square root of the cable length l, i.e., $\delta \propto \sqrt{l}$, where $l = 8.0 - h$, h is the gripper height (meters). Fitting recorded trajectories with principal-component analysis yields the height-dependent safety margin:

$$\epsilon(h) = k\,(8.0 - h)^{0.5} + b, \tag{2}$$

with k and b obtained from the vibration-fitting study in subsection 5.1. Because raw PLC variables do not encode instantaneous swing, the collision checker inflates the fuel-assembly bounding volume by $\epsilon(h)$ before testing against surrounding geometry, thereby reflecting on-site safety envelopes in the DT.

Adaptive Speed Control The simulation advances component transforms by discrete display integration. For a step from x_k to x_{k+1} over Δt, the physics engine generates an interpolated sequence $S_{x_k} = \mathrm{interp}(x_k, x_{k+1}, \Delta t)$ with $|S_{x_k}| = f\,\Delta t$ at frame rate f. Offline runs driven by procedural files lack timing between anchors x_s and x_t, so we reconstruct temporal states with rule-based speeds derived from site drawings. Motions outside constrained areas use $v_{\mathrm{high}} \approx 0.20 \pm 0.05\,\mathrm{m/s}$; slow zones use $v_{\mathrm{low}} = 0.10\,\mathrm{m/s}$. As shown in Figure 7(a), the slow-speed region $\mathcal{Z}_{\mathrm{slow}}$ is dilated by the swing margin $\epsilon(h)$ to form $\mathcal{Z}_{\mathrm{slow}} \oplus \epsilon(h)$; the plant boundary $\mathcal{Z}_{\mathrm{bd}}$ and its dilation $\mathcal{Z}_{\mathrm{bd}} \oplus \epsilon(h)$ serve as emergency-stop and warning regions, respectively. When crossing a high/low boundary, we model joystick latency ($\approx 0.5\,\mathrm{s}$) with bounded acceleration $0.2\,\mathrm{m/s^2}$ until the target speed is reached. For a sampling density γ, anchors are linearly subdivided into $\{x_i\}_{i=0}^{\gamma}$ and each segment duration is set by its target speed v_i:

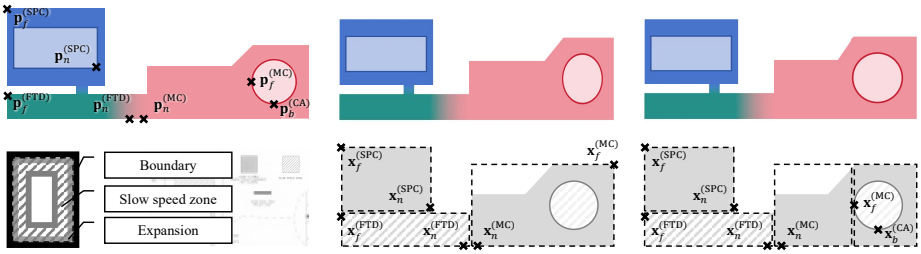
$$S_{x_i} = \mathrm{interp}\left(x_i, x_{i+1}, \frac{\|x_{i+1} - x_i\|}{v_i}\right), \tag{3}$$

then concatenated to yield the full trajectory. Warning regions execute at v_{low} with alerts; emergency regions halt the run with abnormal-state feedback.

Multi-Stage Motion Retargeting In VR, the *translation gain* is:

$$g_t = \frac{\|\Delta\mathbf{X}\|}{\|\Delta\mathbf{P}\|}, \tag{4}$$

where $\Delta\mathbf{P}$ is a displacement in the physical plant and $\Delta\mathbf{X}$ the corresponding displacement in the virtual space. Adjusting g_t allows spatial scaling within perceptual limits. Our multi-stage retargeting maps heterogeneous PMC layouts to a common virtual environment by dividing the plant into semantic regions,



(a) Physical plant anchors and abstracted operation regions.

(b) Naive mapping strategy.

(c) Mapping strategy with critical area consideration.

Fig. 7. Illustration of adaptive speed control zones and facility size retargeting. The diagrams show physical plant anchors, operational regions, and corresponding virtual plant anchors under different mapping strategies. In the naive mapping approach, distortion occurs in the pressure vessel area, whereas incorporating critical area constraints eliminates the distortion. The operational zones shown are illustrative abstractions and do not represent the actual zoning specifications defined in proprietary plant design.

each with physical anchors $\mathbf{p}_n^{(i)}, \mathbf{p}_f^{(i)}$ and virtual anchors $\mathbf{x}_n^{(i)}, \mathbf{x}_f^{(i)}$:

$$\mathbf{g}_t^{(i)} = \frac{\mathbf{x}_f^{(i)} - \mathbf{x}_n^{(i)}}{\mathbf{p}_f^{(i)} - \mathbf{p}_n^{(i)}}, \tag{5}$$

and

$$f_i(\mathbf{p}) = \mathbf{x}_n^{(i)} + \mathbf{g}_t^{(i)} \odot \left(\mathbf{p} - \mathbf{p}_n^{(i)}\right), \tag{6}$$

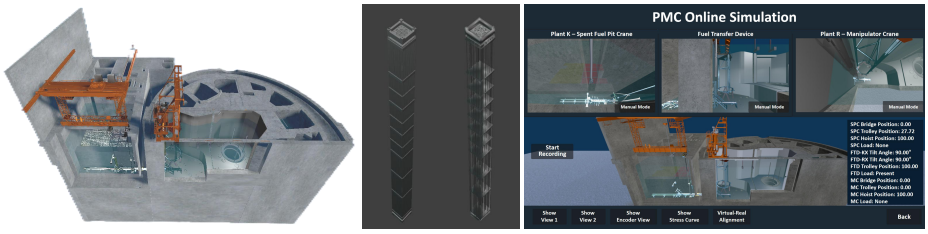
with $\odot$ the Hadamard product. In the core pit, naive near–far mapping distorts the cylindrical pressure vessel, misaligning feed points and causing pseudo-interpenetration in the DT. To fix this, *critical anchors* $(\mathbf{p}_b^{(\mathrm{CA})}, \mathbf{x}_b^{(\mathrm{CA})})$ are added. Outside the critical zone, Equation 6 applies as usual; inside, a strict 1:1 mapping preserves geometry integrity (see Figure 7).

4 Interactive Graphical Interface

The interactive graphical interface serves as the primary human–computer interaction layer of the 3D simulation platform, integrating scene visualization, gesture-based interaction, and multi-perspective camera control. It enables users to intuitively access and manipulate simulation in both online and offline modes.

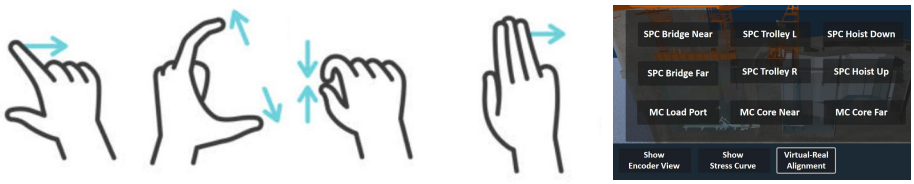
4.1 Scene Visualization and Gesture Interaction

The 3D scene is reconstructed from CAD models provided by the PMC plant owner, covering both civil structures and major equipment layouts. Original CAD drawings are converted into `.fbx` format via a standardized processing



(a) Geometry overview. **(b)** Fuel LOD. **(c)** Interface layout.

Fig. 8. Examples of 3D asset rendering details and an overview of the online simulation visualization interface. (b) shows the full rendered fuel on the left, with 10% simplified meshes for static elements rendering on the right.



(a) Panning. **(b1)** Zoom in. **(b2)** Zoom out. **(c)** Camera rotation. **(d)** VR anchor alignment.

Fig. 9. Three manual touch-based camera interaction modes and the virtual–real anchor alignment interface.

pipeline, preserving geometry hierarchies and texture mapping to ensure accurate digital representation. The finalized plant assets are shown in Figure 8(a).

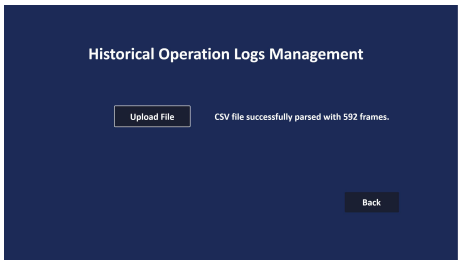
To enhance spatial comprehension and maintain rendering efficiency, the visualization system integrates the following features:

- **Level of Detail (LOD) rendering**: As shown in Figure 8(b), static fuel elements in the pits are rendered with simplified meshes, reducing polygon count and improving frame rates without sacrificing operational context.
- **Section view rendering**: Enables cross-sectional cuts of buildings and equipment to reveal internal structures for inspection and training in real-time.
- **Surface for water bodies**: Simulates dynamic water surfaces using a spring system to provide realistic depth cues and physical responses to interactions.
- **Multi-camera architecture**: Comprises three interactive subsystem cameras of SPC, FTD, and MC, along with a fixed global main overview camera for continuous scene monitoring.

Gesture-based interaction is implemented for intuitive control on touch-enabled devices (Figure 9(a–c)). Users can manually pan, zoom, and rotate the viewport, select interactive components, switch to a director mode for automatic tracking of objects of interest, or perform keyword-based searches to focus on specific components with detailed close-up views. Before system operation, to ensure proper visualization without model interpenetration, users must move the gripper to a



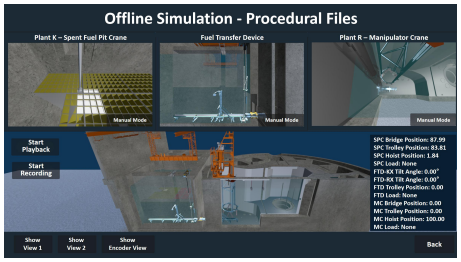
(a) Initialization interface.







(b) Offline simulation with playback interface.



(c) Offline simulation with procedure interface.

Fig. 10. Interface layouts for key platform functions, including initialization, historical log playback, and procedural file-based offline simulation.

designated real-world position and perform manual virtual–real alignment using the anchor alignment interface shown in Figure 9(d).

4.2 Functional Layout of the Interface

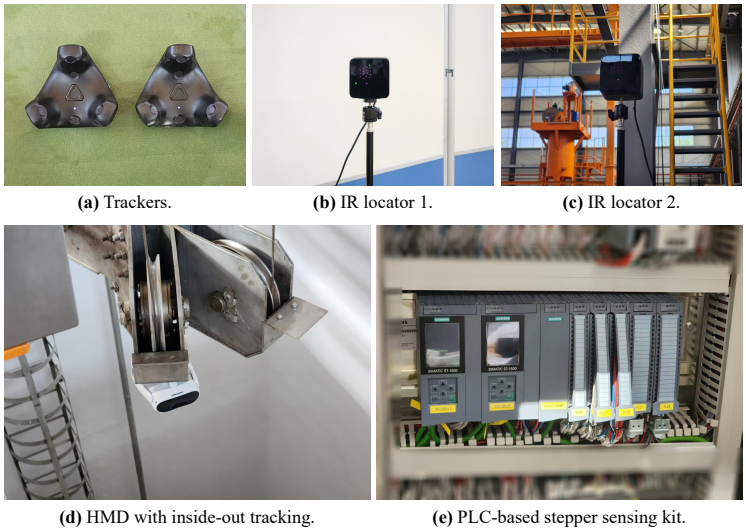
The functional layout of the interface, shown in Figure 10, comprises a login screen, a main menu, and dedicated views for each operational mode:

- **Initialization interface:** Handles user authentication and role-based access control, ensuring secure operation and mode management (Figure 10(a)).
- **Online simulation interface:** Displays real-time PMC equipment status synchronized with physical devices, supporting live simulation (Figure 8(c)).
- **Playback interface:** Enables loading and replaying of historical operation logs for offline simulation (Figure 10(b)).
- **Procedure interface:** Supports loading, editing, training, and execution of procedural files for offline simulation (Figure 10(c)).

5 System Validation

This section presents the experimental validation of the proposed system, focusing on the gripper vibration fitting process and the analysis of system-level errors. All tests were conducted under realistic operating conditions to ensure the results are representative of actual PMC refueling scenarios.

5.1 Gripper Vibration Fitting Evaluation



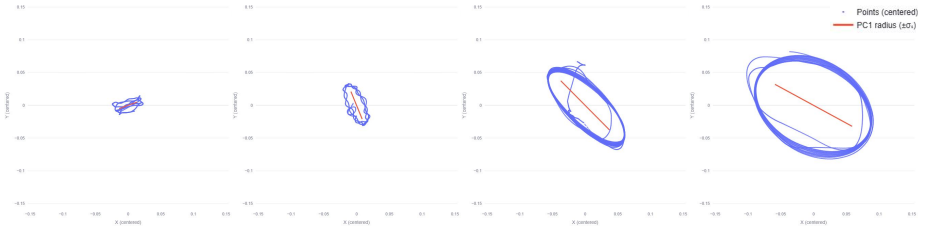
(a) Trackers. **(b)** IR locator 1. **(c)** IR locator 2.

(d) HMD with inside-out tracking. **(e)** PLC-based stepper sensing kit.

Fig. 11. Overview of equipment used for on-site experimental data acquisition.

To characterize the oscillatory behavior of the gripper and to determine collision-related parameters for subsequent error analysis, we recorded motion trajectories under a range of operating conditions. Two complementary acquisition methods were used: (1) optical positioning with the *tracker* setup shown in Figure 11(a–c); and (2) a novel approach in which an XR headset with *inside-out tracking* capability (Figure 11(d)) was physically mounted on the gripper to record trajectories using its onboard IMU and visual–inertial fusion algorithms. The latter method effectively addressed the limitation of conventional trackers, which cannot capture trajectories inside the pit at ground level (0 m), thereby enabling complete motion data coverage. In addition, the central PLC control unit illustrated in subsection 3.1 has its corresponding stepper sensing kit shown in Figure 11(e).

Trajectory data were collected at a maximum rate of 60 Hz, theoretically resulting in more than 120,000 points spanning oscillations at multiple heights. The test procedure was as follows: the gripper was positioned at a specified height,

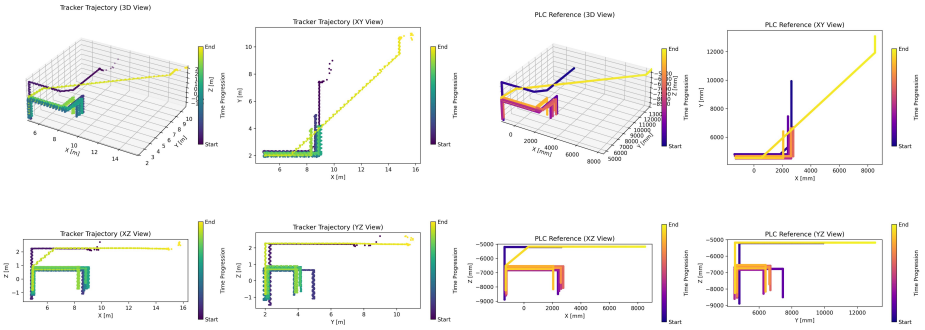


(a) Gripper at height 3m. **(b)** Gripper at height 1m. **(c)** Gripper at height -1m. **(d)** Gripper at height -3m.

Fig. 12. End-effector oscillation trajectories (blue) of the gripper after emergency stops at different heights, with PCA results (principal axis σ_1 in red). The ground level is taken as the 0 m reference.

the crane's trolley and bridge were accelerated to maximum speed, and then brought to an abrupt stop. The subsequent free-swing trajectory was recorded in its entirety. Representative results are shown in Figure 12. For each height, principal component analysis (PCA) was applied to extract the swing eccentricity. The primary axis length σ_1 was taken as the target ϵ for curve fitting, in accordance with Equation 2. The fitted parameters were $k = 0.0882$ and $b = -0.1721$, yielding the functional relationship $\epsilon(h)$. This model is directly integrated into the simulation platform's collision-warning module to enable early prediction and risk assessment of potential interference events.

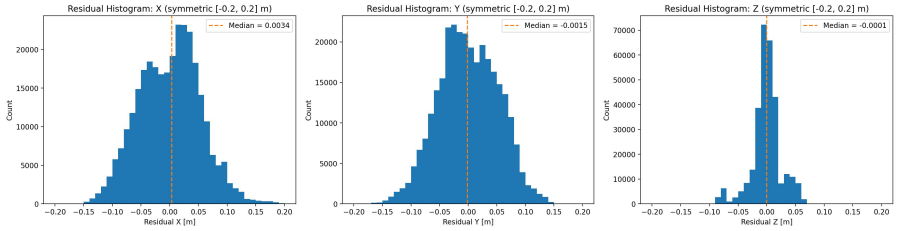
5.2 System Error Analysis



(a) Measured trajectory from tracker-mounted gripper. **(b)** Reference trajectory encoded by the system PLC.

Fig. 13. Pseudo-color visualization of the gripper end-effector trajectory over time, comparing (a) tracker-based measurements and (b) PLC-encoded reference data.

Building on the tracking strategy in subsection 5.1, we recorded the gripper's zero-point motion trajectory during continuous fuel-loading operations of



(a) X-axis residual distribution. **(b)** Y-axis residual distribution. **(c)** Z-axis residual distribution.

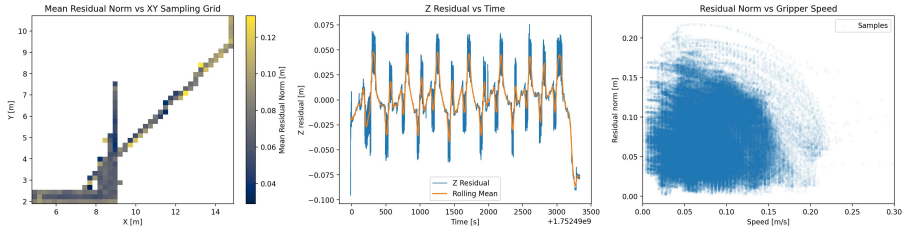
Fig. 14. Statistical distributions of system residuals along the X, Y, and Z axes.

the PMC system to analyze the distribution and influencing factors of system error. The dataset covers multiple consecutive loading sequences over a one-hour period. After 10× uniform downsampling, Figure 13(a) shows the ground-truth trajectory from the tracker, while Figure 13(b) shows the corresponding PLC-encoded trajectory retrieved from the Equipment Data Pool System. Qualitatively, the trajectories align closely, indicating consistent motion between the physical and virtual environments, and providing preliminary validation of the 3D simulation platform.

To quantify the effect of fine-scale motion jitter and mechanical characteristics, residuals were computed between the two trajectories. After converting encoder outputs from millimeters to meters and applying a uniform coordinate bias correction, the median residuals in x, y, and z were $m_x = 0.0034$, $m_y = -0.0015$, and $m_z = -0.0001$, respectively. The histogram of these residuals is shown in Figure 14. Mean residuals for all components were below the vertical motion tolerance of 0.005 m specified in the hardware architecture (see subsection 3.1). A joint t-test over all three components yielded $p = 0.9954$, indicating no statistically significant deviation from zero and thus no evident cumulative systematic error. The absolute residual values were $a_x = 0.0438 \pm 0.0537$, $a_y = 0.0428 \pm 0.0523$, and $a_z = 0.0177 \pm 0.0254$, all within the 0.05 m cumulative error tolerance defined in the software architecture. Skewness analysis showed $\gamma_x = 0.070$ and $\gamma_y = -0.036$, both below the industrial threshold of 0.1, indicating good symmetry and minimal tail bias. For z, $\gamma_z = -0.561$ suggested mild negative skewness, likely caused by slight cable elongation under fuel load, manifesting as a small downward drift when loaded.

Table 1. Summary of residual statistics over segmented speed bins.

Speed bin	Count	Residual							
		R_n $(abs \downarrow)$	R_n^{max} $(\downarrow)$	R_x $(abs \downarrow)$	R_x^{max} $(\downarrow)$	R_y $(abs \downarrow)$	R_y^{max} $(\downarrow)$	R_z $(abs \downarrow)$	R_z^{max} $(\downarrow)$
$[0.00, 0.05)$	71887	0.0708 ± 0.0335	0.2165	0.0024 ± 0.0508	0.2031	-0.0025 ± 0.0540	0.1658	-0.0031 ± 0.0250	0.0902
$[0.05, 0.10)$	161679	0.0707 ± 0.0340	0.2171	-0.0016 ± 0.0534	0.2062	$\mathbf{-0.0017} \pm 0.0510$	0.1794	$\mathbf{0.0006} \pm 0.0263$	0.0903
$[0.10, 0.15)$	59459	0.0750 ± 0.0356	0.2136	$\mathbf{0.0001} \pm 0.0579$	0.2054	0.0093 ± 0.0537	0.1702	0.0025 ± 0.0237	0.0950
$[0.15, 0.20)$	7391	$\mathbf{0.0646} \pm 0.0343$	0.1924	0.0050 ± 0.0543	0.1900	-0.0081 ± 0.0433	0.1374	-0.0015 ± 0.0210	0.0955
$[0.20, 0.25)$	894	0.0702 ± 0.0237	0.1619	0.0421 ± 0.0421	0.1305	-0.0340 ± 0.0229	0.0814	-0.0015 ± 0.0163	0.0874
$[0.25, 0.30)$	12	0.0847 ± 0.0305	**0.1226**	0.0537 ± 0.0214	**0.0948**	-0.0388 ± 0.0304	**0.0631**	-0.0412 ± 0.0274	**0.0849**



(a) Error spatial pattern. **(b)** Error temporal evolution. **(c)** Error relationship with speed.

Fig. 15. Spatiotemporal characterization of system errors, covering spatial pattern, temporal evolution, and speed correlation.

We further analyzed the spatio-temporal characteristics of the residuals. Figure 15(a) shows a top-view heatmap of mean residual norms, highlighting concentration in critical areas such as the spent fuel rack and connecting gate, with larger deviations in unloaded zones (e.g., near cell (14, 8)). Figure 15(b) plots the z-axis residual time series over the one-hour run, which included seven loading events. The residual exhibits a periodic rise–fall pattern, with peaks largely within the 0.05 m design tolerance; this periodicity is consistent with load-induced cable stretching and oscillations. Finally, Figure 15(c) shows residual magnitude versus velocity, obtained by differentiating trajectories by timestamp. The $[0, 0.3)$ m/s velocity range was discretized into 0.01 m/s steps. Table 1 futher summarizes the residual statistics across different speed bins. It reveals an empirical envelope that decreases with speed, suggesting that at higher velocities, anisotropic effects tend to align error vectors with the motion direction, compressing the residual spread.

6 Conclusion, Limitations, and Future Work

This work presented the design, implementation, and validation of a PMC-oriented 3D simulation platform that integrates hardware sensing, PLC-based control, and multi-level software subsystems for quasi-digital twin modeling. The platform enables real-time visualization, procedural offline simulation, adaptive control algorithms, and motion retargeting, while supporting both operator training and system analysis under realistic plant constraints. Comprehensive experiments, including gripper vibration fitting and system-level error evaluation, demonstrated close correspondence between the virtual and physical systems.

Several limitations remain. First, the digital twin representation focuses on geometric and kinematic fidelity rather than full physical-process coupling, limiting its predictive depth. Second, the current implementation does not provide comprehensive on-site command execution; direct control over critical mechanical subsystems is intentionally restricted for safety reasons. Third, long-term drift or load-induced bias in mechanical components is monitored but not yet addressed through automated correction. Finally, the system lacks intelligent data-

driven analytics for long-horizon operational patterns, such as fatigue-induced deviations.

Future work will address these limitations by: (1) extending the DT framework to incorporate higher-fidelity physical and thermal models; (2) enhancing secure bidirectional control channels to support supervised execution of key mechanical actions; (3) developing automated compensation strategies for cumulative drift based on statistical and learning-based approaches; and (4) introducing intelligent analysis modules for predictive maintenance, ergonomic assessment, and multi-scenario optimization of refueling operations.

Acknowledgments. This work is supported by the National Natural Science Foundation of China through Project 61932003, 62372026, by Beijing Science and Technology Plan Project Z221100007722004, by National Key R&D plan 2019YFC1521102, and by the fundamental research funds for the central universities.

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